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Cross-attention driven RGBD fusion in lightweight CNN–Transformer for real-time stress classification of seedlings during robotic transplanting

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ABSTRACT

Timely selection of high-quality seedlings during robotic transplanting is crucial for ensuring optimal crop growth and productivity. However, accurate real-time seedling grading is technically challenging due to subtle visual differences among stress levels and the limited computational capacity of edge devices. To address this, we propose a lightweight RGBD (RGB and depth) multimodal framework for six-level stress classification of bell pepper seedlings. Our approach integrates a dual-stream CNN–Transformer architecture in which hierarchical cross-attention modules fuse color-texture cues from the RGB stream with depth-derived geometric cues at both shallow and deep feature layers. The compact tokenizer rapidly extracts multi-scale fused features, which allow transformer encoders to capture long-range dependencies, and an attention-pooling head produces class logits. To validate our algorithm, a dedicated dataset of 1537 paired RGBD images was collected with an oblique-view sensor during the pick-and-place cycle. Deployed on an edge-device, the network achieves an overall accuracy of 95.8 % and maintains 5.2 frames per second when integrated into the parallel transplant-control pipeline. Ablation studies reveal that effective fusion of RGBD channel can improve accuracy by 4.9 % over RGB-only baseline model, which supports the effectiveness of the modality-aware design. This study integrates RGBD imaging with an edge-ready hybrid network to provide a practical tool for automated seedling transplanting and an effective solution for multimodal perception in precision agriculture.

1. Introduction

Greenhouse cultivation has expanded rapidly since the 2000s, covering at least 1.3 Mha as of 2019, and continues to grow globally, driven by the demand for stable crop production in controlled environments (Tong et al., 2024). In parallel, the adoption of agricultural robotics has accelerated across various stages of the crop lifecycle, including seedling cultivation, transplantation, spraying, and harvesting. Among these, transplanting robots have received particular attention for their potential to improve labor efficiency and operational precision. Transplantation is widely recognized as a critical agronomic intervention, as the selection and placement of seedlings at this stage significantly influence subsequent crop growth, resource uptake efficiency, and ultimately final yield. Traditional manual transplantation relies on the experience of farmers or agricultural workers, but the process is labor-intensive, time-consuming, and often inconsistent, underscoring the practical demand for automated solutions (Otoya and Gardini, 2021). The emergence of seedling transplanting robots marks a significant advancement in automation, substantially reducing labor demands and operational time. Improved transplanting efficiency was achieved by enhancing the precision of robotic arms, underscoring the potential of such systems to replicate accurate human actions (Zhao et al., 2023). However, these robots cannot assess seedling quality, which can affect crop performance and yield. Developing a system capable of effectively and accurately evaluating seedling quality remains a critical challenge with significant practical implications.

A variety of approaches have been explored for assessing seedling quality, including seedling stress detection and seedling health assessment. Early research on seedling quality classification primarily relied on hyperspectral and thermal imaging techniques (Lowe et al., 2017). These techniques effectively extracted plant quality indicators based on spectral signatures at specific wavelengths. While these modalities provide fine-grained insights into plant biochemistry, they face

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several practical limitations. High equipment costs and complex calibration procedures hinder field scalability, while the resulting high-dimensional data demand computationally intensive preprocessing and dimensionality reduction using PCA and PLSR (Sarić et al., 2022). These factors collectively limit their real-time applicability and robustness in dynamic agricultural environments. With the development of machine learning, algorithms such as support vector machine and k-nearest neighbors have been increasingly applied to process agricultural image data (Attri et al., 2024). Although these methods are more efficient than early statistical methods for processing data, they have limitations in multi-class classification tasks and are sensitive to parameter settings, which can affect their performance under different conditions. More importantly, the accuracy of traditional methods heavily relies on carefully designed handcrafted feature descriptors, making it challenging to select the optimal and robust features from a large pool of extracted features. Consequently, traditional machine learning techniques have shown limited adoption in large-scale agricultural production (Kok et al., 2021).

With the rapid development of deep learning, especially the advent of convolutional neural networks (CNNs), significant progress has been made in the detection of seedling quality status. CNNs, by increasing the depth of the network and applying various convolutional operations, can hierarchically represent data. This capability makes CNNs particularly well-suited for learning local features and hierarchical representations in images (Alzubaidi et al., 2021). In diverse agricultural applications, several classic CNN-based models have been extensively employed for disease classification and quality assessment (Samiei et al., 2020). However, CNN architectures struggle to capture long-range dependencies and the global structure of images. When processing images with scattered features, it is difficult for CNNs to effectively identify global context and spatial relationships. Furthermore, CNNs often involve a large number of parameters, particularly in deeper networks (Alzubaidi et al., 2021).

In recent years, vision transformers have emerged as a promising alternative that can improve accuracy and learning capabilities. These models represent a shift from traditional CNN methods and focus on capturing global context and relationships within image data (Han et al., 2023). Building on this foundation, many high-performing models, such as swin transformer (Liu et al., 2021), have been proposed and have achieved excellent performance in various computer vision tasks. However, despite their improved accuracy, transformer-based image recognition models still have the drawbacks of being computationally extensive and architecturally complex. For small datasets, transformer architectures perform poorly in the absence of multi-scale feature maps and robust local feature extraction capabilities (Han et al., 2023).

Accumulated literature evidence indicates that RGB-based approaches have achieved notable advances; nevertheless, their single-modal nature inherently limits the information captured, restricting models from representing comprehensive plant traits (D. Li et al., 2020). Regarding the assessment of seedling quality, analyzing RGB images of seedlings can help determine their health status. Healthy plants typically exhibit vibrant green colors, while unhealthy or disease-affected plants appear yellowish or have spots (Agarwal and Dutta Gupta, 2018). In addition, depth information can provide valuable scale cues for analyzing the 3D structure of plants, including leaf size, shape, and overall plant morphology, which are equally important for assessing seedling quality (Z. Li et al., 2020). Multimodal data exhibits both complementarity and consistency, allowing the modalities to compensate for each other's limitations and deficiencies. As a result, the perception and interpretation of multimodal information have become a prominent research focus. For example, Sun et al. (2022) proposed an attention-based fusion module to effectively integrate RGBD (RGB+depth) information, mitigating the adverse effects of depth noise, and significantly improving detection accuracy. In Xu et al. (2021), a novel approach was introduced in which single-channel depth images were re-encoded into three-channel images resembling RGB structures. The subsequent integration of these re-encoded depth images with RGB data resulted in a significant improvement in weed detection accuracy.

While recent progress in plant phenotyping and vision-based classification has significantly advanced the field, many existing approaches remain constrained by their reliance on unimodal data sources. In particular, they often use RGB imagery or vegetation indices derived from limited spectral bands. These methods primarily focus on observable surface characteristics such as color intensity, leaf area, plant height, or chlorophyll content. Although such features provide valuable visual cues, they not fully capture subtle physiological stress indicators or underlying morphological traits that are critical for determining transplant readiness from an agronomic perspective. Moreover, such parameters are often insufficient to detect early-stage physiological stress, structural anomalies, or internal health conditions that are vital for predicting post-transplant survival. Furthermore, these systems generally lack the capacity to jointly model surface texture and volumetric cues, which are essential for evaluating key morphological indicators such as seedling compactness, leaf curling, stem straightness, and root-substrate integration. In automated transplanting scenarios, robotic grippers must rapidly manipulate and assess individual seedlings with sub-second latency. Consequently, classification models must operate with real-time responsiveness and detect subtle quality variations that are often imperceptible in 2D RGB imagery. While hyperspectral imaging and full-resolution 3D reconstruction offer rich phenotypic detail, their high equipment costs and computational demands render them impractical for deployment on memory- and power-constrained edge devices. It is beneficial to have a lightweight, multimodal classification framework capable of real-time operation on low-power platforms, with robust cross-modal feature representation and fusion capabilities.

In response to the practical demands of real-time seedling assessment in automated transplanting systems, we present a task-specific visual classification framework that leverages a CNN–Transformer hybrid architecture to fuse RGB and depth modalities. Our approach is specifically designed to operate under the constraints of an edge-ready embedded platform, enabling accurate, efficient, and real-time evaluation of seedling stress conditions during robotic handling. The specific objectives of this study are as follows: (1) To curate a novel stress-level dataset for bell pepper seedlings, capturing both RGB and depth images in real-time during the seedling transplantation process; (2) To develop an optimized CNN–Transformer hybrid architecture that synergistically combines convolutional feature extraction and attention mechanisms to enhance training efficiency, improve predictive performance, and enables computational efficiency suitable for deployment on resource-constrained edge-device in agricultural environments; and (3) To propose a lightweight cross-attention (CA) multimodal fusion module that effectively integrates RGBD data, significantly improving classification accuracy while meeting stringent computational constraints.

2. Materials and methods

2.1. Experimental setup

2.1.1. Preparation of seedling samples

The design of the dataset is crucial to ensuring the validity and robustness of model training outcomes. In agricultural research, particularly in seedling cultivation, one persistent challenge lies in the limited availability and ambiguous classification of seedlings exhibiting suboptimal growth. Variations in seedling quality are often driven by genotypic differences, as well as environmental factors such as uneven water availability and nutrient uptake inefficiencies (Langenfeld et al., 2022). Among these, water stress and nutrient stress are widely recognized as two of the most critical abiotic factors affecting early-stage plant development, influencing photosynthetic capacity, root architecture, and overall vigor. To systematically investigate the effects of these stressors and generate phenotypically diverse samples, we applied a controlled cultivation protocol that precisely regulated

Table 1
Watering and nutrient supply intervals for the six stress treatment conditions.

	W2N2	W2N0	W4N4	W4N0	W6N6	W6N0
Watering interval (day)	2	2	4	4	6	6
Nutrient supply interval (day)	2	None	4	None	6	None

watering frequency and nutrient supply levels. This strategy enabled us to induce a continuous spectrum of seedling stress levels under reproducible conditions. It also allowed us to construct a well-balanced dataset capturing a wide range of morphological and physiological stress states. The resulting dataset enhances both the statistical reliability and diversity of representational needed for training robust classification models (Li et al., 2022).

We used the plug tray method for seedling cultivation, specifically employing 36-hole trays filled with coconut coir—a widely used soilless substrate known for its good air permeability and water retention. The seedlings were grown following standard cultivation procedures. The cultivation environment was maintained in an indoor incubator with a controlled temperature of 27 °C, relative humidity set at 70 %, and 15 h of simulated daylight provided each day. The water supply regimen was adjusted according to growth stages: 1 L per application during the early stage (0–15 days), 1.5 L during the mid-stage (15-30 days), and 2 L during the late stage (30–45 days). The nutrient solution was prepared using a water-soluble nitrogen–phosphorus–potassium compound fertilizer (20–20–20). During the early stage (0–15 days), seedlings were irrigated with a 0.25 g/L solution (i.e., 1/4 strength). From day 15 onward, the concentration was increased to 1.0 g/L (full strength).

Table 1 presents the experimental groups along with their respective water and nutrient supply regimens. The control group, denoted as W2N2, received a standard regimen of water and nutrients, which was quantified and administered every 2 days, as the baseline condition. The experimental treatments were designed to simulate reduced absorption capacity in seedlings. This was achieved by modulating the frequency and volume of inputs. As a result, the dataset was enriched with samples representative of lower quality. For instance, to cultivate seedlings of diminished quality, the frequency of water supply was reduced (e.g., once every 6 days for the W6N0 group) in combination with the suspension of nutrient supply, thereby enhancing the representational diversity of these samples within the dataset.

This strategic cultivation approach enabled the efficient assembly of a diverse seedling stress dataset and additionally provided the machine learning models with a wealth of balanced and informative training data. Such a dataset augments the models' generalization capabilities and predictive accuracy.

2.1.2. Embedded platform and RGBD imaging system

As shown in Fig. 1a, to ensure reliable depth acquisition, the camera-to-subject distance was fixed at 30 cm, slightly exceeding the minimum depth sensing range of 25 cm specified by the hardware. During transplantation, seedlings were grasped by a custom-designed gripper with a minimized contact area, allowing root visibility without compromising mechanical stability. Importantly, imaging occurs after the seedling is lifted from the tray, enabling per-plant data collection without occlusion from neighboring seedlings. A uniform background or depth-based segmentation technique is applied to isolate the target seedling, ensuring clean and consistent inputs for subsequent analysis.

The hardware configuration of the seedling transplant system is built using the Jetson AGX Xavier (NVIDIA Corporation, Santa Clara, CA, USA), an embedded AI platform equipped with an 8-core ARM CPU and 512-core CUDA GPU, capable of delivering up to 32 TOPS for real-time inference, as shown in Fig. 1b. This platform controls the robotic arm and force-feedback gripper for precise transplant operations, while simultaneously acquiring RGB and depth images using the Azure Kinect DK RGBD camera (Microsoft Corporation, Redmond, WA, USA). The force-feedback mechanism is incorporated to accommodate variations in root–substrate block size among seedlings. By regulating gripping force, it ensures stable lifting without damaging the root substrate, thereby maintaining consistent imaging conditions for RGBD perception. The camera captured synchronized high-resolution RGB images (4,096 × 3072 pixels) and aligned depth maps (640 × 576 pixels) at 5 frames per second, thereby enabling real-time seedling stress assessment. The integrated control interface supports unified management of all modules, thereby facilitating automated data acquisition and transplant execution.

Fig. 1c depicts the optimized camera mounting configuration set at a 35° oblique angle, which was determined through kinematic simulation and empirical tuning. This placement facilitates the simultaneous capture of key seedling organs (leaves, stems, root-substrate blocks), thereby enhancing the completeness of morphological information. Fig. 1d depicts a conceptual schematic of the anticipated field-deployment platform, illustrating the envisioned transplanting workflow in a greenhouse setting. In this scenario, a 6-DoF robotic arm transfers seedlings from a high-density nursery tray to a low-density tray, demonstrating the potential operational process of the transplanting robot system under real-world cultivation conditions. The overall control process follows a perception–decision–execution loop. Specifically, after a seedling is grasped and lifted by the robotic gripper, synchronized RGBD images are acquired and processed on the Jetson AGX Xavier. The proposed classification network performs real-time stress-level prediction. Based on the inferred stress category, the controller determines the subsequent transplant action. This closed-loop design enables seamless integration of visual perception and robotic control during the transplanting process.

2.1.3. RGBD seedling dataset

A representative subset of the constructed RGBD seedling dataset is illustrated in Fig. 2, featuring seedlings from six distinct stress treatment groups after 45 days of cultivation. Each subfigure presents the RGB image (left) and its corresponding depth image (right), offering a dual-modality visualization of seedling morphology. The visual diversity across groups is a direct consequence of the differential water and nutrient regimens outlined in Table 1, and reflects variations in physiological responses such as leaf area, extent of chlorosis, stem thickness, and root plug development.

Specifically, seedlings in the W2N2 group (Fig. 2a), which received optimal watering and nutrient input, exhibit vibrant green leaves with broad surfaces and dense canopy structures that are indicative of robust growth. Conversely, seedlings in the W6N0 group (Fig. 2f), subjected to severe abiotic stress, display prominent stress symptoms, including stunted leaf growth, chlorotic discoloration, and reduced foliage density. Intermediate groups (e.g., W4N4 and W4N0) show more subtle phenotypic differences, such as slight yellowing or marginal reduction in leaf area, which are visually less discernible and present significant challenges for classification models. This highlights the necessity for fine-grained feature extraction and robust multimodal fusion to accurately capture such nuanced morphological indicators.

To improve generalization and enhance model robustness under diverse visual conditions, data augmentation techniques were applied to the RGB images for training dataset. These techniques included horizontal flipping and random adjustment to brightness and contrast within a range of [0.8, 1.2], simulating varying lighting environments. The aligned depth map was merely flipped horizontally whenever its paired RGB image was flipped; no brightness or contrast operations

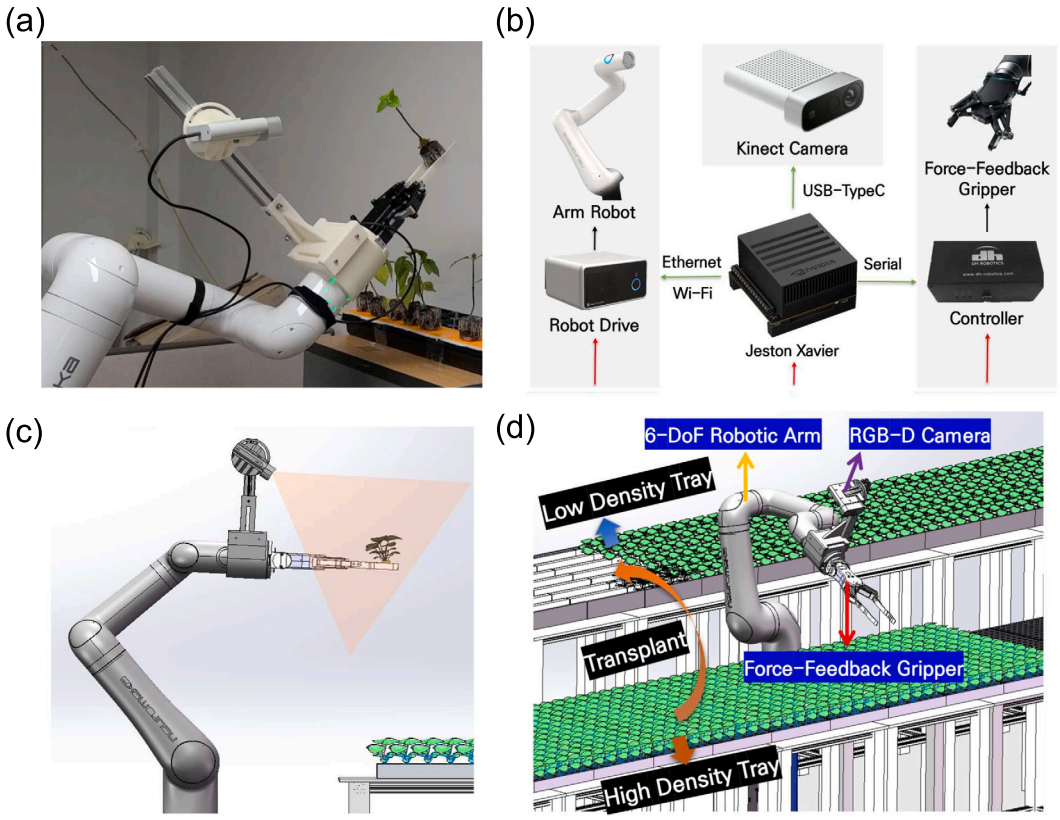


Fig. 1. Transplanting robot system: (a) image acquisition of seedlings using the robot platform, (b) configuration of the embedded robot control system, (c) schematic of the camera and its field-of-view, and (d) conceptual scenario illustrating potential future field deployment of the transplanting robot system in a greenhouse environment.

Table 2
Number of original RGBD images for training and testing.

Group	Training	Testing	Total
W2N2	230	57	287
W2N0	215	53	268
W4N4	225	56	281
W4N0	195	48	243
W6N6	204	51	255
W6N0	163	40	203
Total	**1232**	**305**	**1537**

were applied to depth data so that pixel-wise geometry remained unchanged. These augmentations effectively doubled the size of the training dataset without altering its semantic content, thereby enriching feature diversity and reducing the risk of overfitting.

A total of 2769 image samples were utilized in this study, comprising 1537 original images collected through field acquisition and 1232 generated images using data augmentation. For each experimental group, the dataset was randomly partitioned into training and testing subsets using an 8:2 split ratio, as shown in Table 2. This division ensures that the model's performance can be adequately evaluated on different datasets while preserving a sufficient amount of independent data to assess the model's generalization ability.

2.2. Network architecture

In this section, we systematically presented how to effectively utilize multimodal data (color and depth) to improve the accuracy of seedling stress classification, starting from the basic structure. This section included the RGB-CTA (Convolution–Transformer–Attention) network, which uses only RGB images; the RGBD-CTA-D (Data Layer Fusion) network, which combines RGB and depth data directly at the input layer; and the RGBD-CTA-F (Feature Layer Fusion) network, which fuses RGB and depth features using CA mechanisms. These approaches explore different strategies for integrating color and depth information to enhance the classification accuracy of seedling stress.

2.2.1. Baseline model: RGB-CTA network

We presented a detailed overview of a novel network that combines CNN and transformer architectures with the aim of improving the accuracy and efficiency of seedling stress classification. The network architecture is summarized in Fig. 3 and consists of two key components: tokenizer and classifier.

The tokenizer is responsible for rapidly extracting rich feature maps from the input RGB images using the CRP (Convolutional, ReLU, Pooling) module. Subsequently, to adapt to the transformer module, the feature maps are flattened into one-dimensional vectors and encoded with positional embeddings. This step allows the model to focus on local information at different positions within the feature maps. The classifier receives the one-dimensional feature vectors with positional embeddings and passes them through the transformer encoder module to extract deep-level local and global information about the seedlings. The output is further refined using the attention pooling module and then classified using a multi-layer perceptron (MLP).

The design of the entire architecture effectively combines the powerful feature extraction capabilities of CNN with the global context understanding abilities of transformer. In the following paragraphs, we will systematically present and discuss the specific design and functionalities of the Tokenizer and Classifier components.

2.2.2. Tokenizer module for feature encoding

A detailed illustration of the tokenizer component, which serves as the initial stage of our CNN–Transformer network designed for

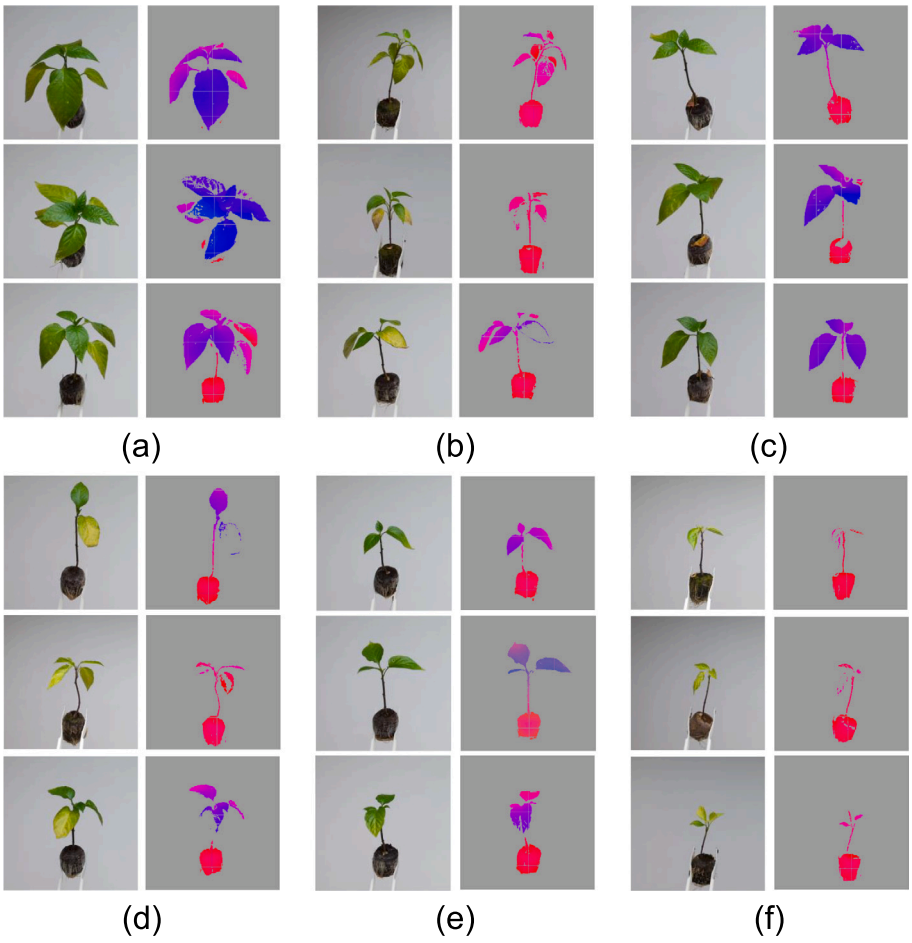


Fig. 2. RGB (left) and depth (right) images of the six experimental groups: (a) W2N2, (b) W2N0, (c) W4N4, (d) W4N0, (e) W6N6, and (f) W6N0.

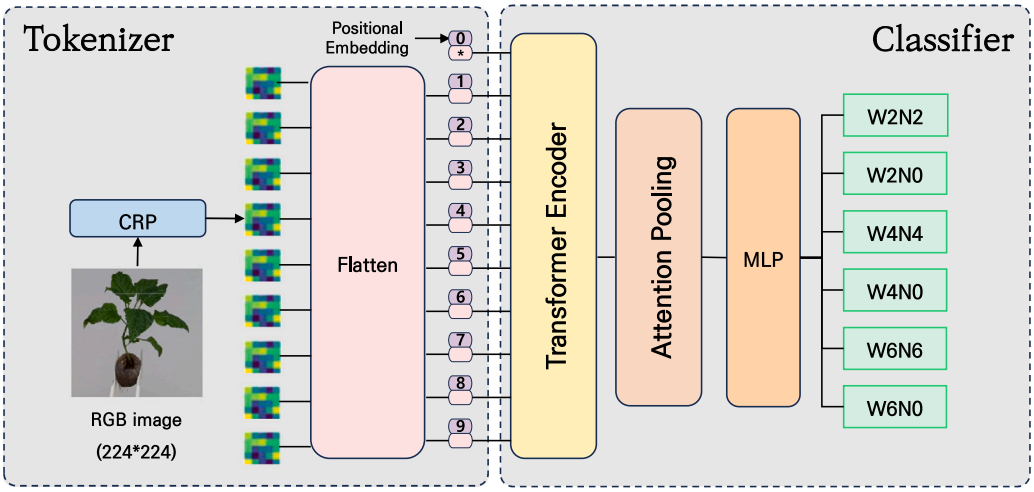


Fig. 3. RGB-CTA classification network architecture based on tokenizer and classifier modules.

seedling stress classification, is presented in Fig. 4. It comprises two CRP modules and a flattening layer.

Each CRP module consists of the following components: a 2D convolutional layer (Conv2d) with a kernel size of 7×7 and a 3×3 padding, a ReLU activation function to introduce non-linearity, and a 2D max-pooling layer (MaxPool2d) with a kernel size of 3 and padding of 1, which reduces the image size by half. CRP[0] and CRP[1] differ in their output channel sizes, which are 64 and 256, respectively. This progressive channel expansion enhances the network's ability to learn hierarchical and semantically rich features.

Let $x \in \mathbb{R}^{H \times W \times C}$ denote the input image or feature map, where H and W represent the spatial resolution, and C denotes the number of input channels ($C = 3$ for RGB or $C = 4$ for RGBD input). The transformed feature map after CRP processing is denoted as z, and is

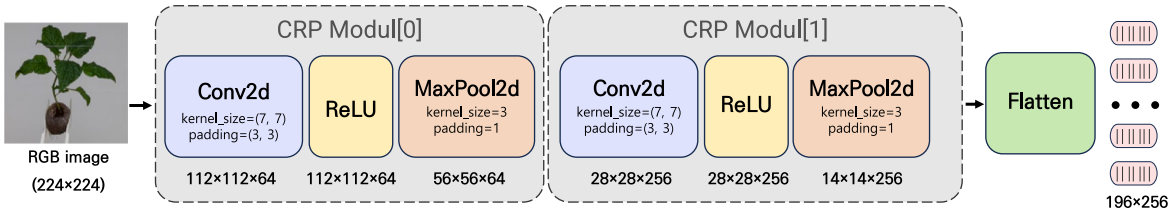


Fig. 4. Tokenizer architecture for extracting initial RGB image features using CRP modules.

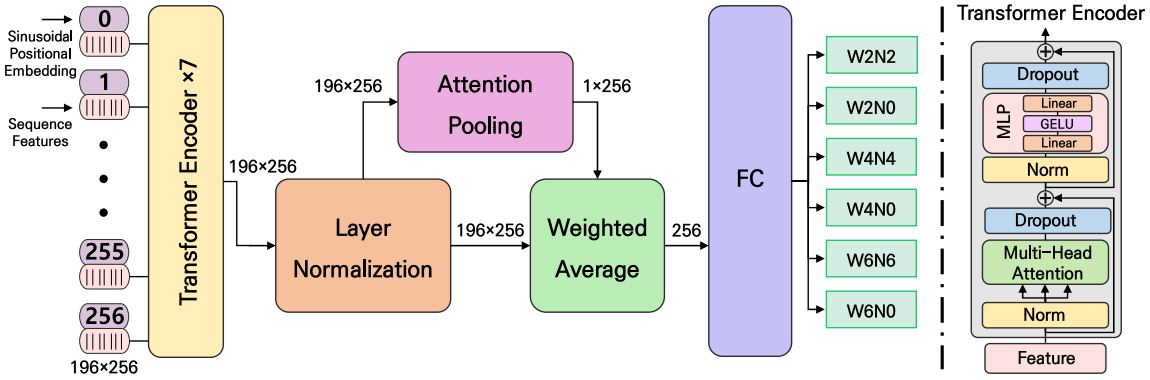


Fig. 5. Classifier module incorporating transformer encoders and attention pooling.

computed as:

$$z = \text{MaxPool}(\text{ReLU}(\text{Conv2d}(x))) \tag{1}$$

After passing through two CRP blocks, the original input of size $224 \times 224 \times C$ is downsampled to $14 \times 14 \times 256$, achieving a spatial reduction of 1/16 while preserving and enhancing salient semantic features. The resulting feature map $z \in \mathbb{R}^{14\times14\times256}$ is then flattened into a sequence of length 196, forming the input for subsequent positional encoding and transformer-based processing.

The design of the tokenizer enables the model to process image data more efficiently and improves sensitivity to critical features while maintaining computational efficiency. This is crucial for the model's ability to accurately classify seedling stress, especially in real-world scenarios where seedling images can be affected by various lighting and background noise. Through such preprocessing and feature extraction, a solid foundation is established for the deeper understanding and accurate classification of seedling stress.

2.2.3. Classifier module and attention-based pooling

A more detailed overview of the classifier component structure is shown in Fig. 5. It consists of 7 transformer encoders, layer normalization, attention pooling, weighted averaging, and a fully connected (FC) layer.

The input to the classification module comprises a sequence of feature vectors output by the tokenizer module, where each vector encodes a distinct local region from the original RGB input. This sequence is represented as a tensor of dimensions 196×256, corresponding to 196 spatial tokens, each with a feature dimensionality of 256. To retain spatial structural information, a sinusoidal positional encoding strategy is applied. This technique assigns a unique position-aware embedding to each token based on its index in the sequence. By incorporating these embeddings, the model can capture not only the semantic content of each region but also their relative spatial arrangement, thereby enabling more contextually aware classification.

Each transformer encoder in the classifier section follows the architectural design of the original vision transformer and transformer (Dosovitskiy et al., 2020), consisting of a multi-head attention (MHA) layer and an MLP block. These blocks are further enhanced with layer normalization, gaussian error linear unit (GELU) activation function, and dropout layers to increase the model's non-linear expressiveness and generalization capability.

The transformer module functions as a sequence-to-sequence model, specifically expressed as two consecutive functions, F_1 and F_2, where F_1 represents the self-attention function, and F_2 denotes the feedforward network function. In detail:

$$F_1(X) = \phi\left(X + \sum_{i=1}^{h} W_0^i \left(W_v^i X\right) \sigma_1\left(\left(W_k^i X\right)^\top \left(W_q^i X\right)\right)\right) \tag{2}$$

$$F_2(X) = \phi\left(F_1(X) + W_2\sigma_2\left(W_1 \cdot F_1(X) + b_1 \mathbf{1}_n^\top\right) + b_2 \mathbf{1}_n^\top\right) \tag{3}$$

Where, X represents the input embedding, ϕ is the layer normalization function, and σ_1 is the softmax function. W_q^i, W_k^i and W_v^i are projection matrices used to generate queries, keys, and values, respectively. W_0^i, W_1 and W_2 are linear transformation weights. b_1 and b_2 are bias terms and σ_2 is the GELU activation function. 1_n denotes a vector of ones, used to broadcast bias terms across sequence elements. Through the self-attention mechanism, the model is able to capture long-range dependencies within the input sequence, enabling it to analyze relationships between elements that are far apart.

Normalization is applied to the output of each sub-layer to stabilize the training process and enhance the model's sensitivity to positional information across the input feature sequence. The outputs of F_1 and F_2 maintain the same dimensions as the input, enabling the stacking of multiple encoder layers to further refine features and enhance representational capacity. Layer normalization is also applied after the transformer encoder to stabilize the network's training process and improve the model's generalization ability.

In the CNN–Transformer network, the attention pooling layer plays an important role by focusing on salient information within the sequence through a weighted allocation mechanism, compressing it into a consolidated representation. This process enables the model to emphasize the most informative features while downplaying less relevant details, thereby improving classification efficiency and accuracy.

Suppose H is the hidden state matrix obtained from the transformer encoder, where $H \in \mathbb{R}^{n\times d}$, n is the sequence length, and d is the feature dimension. Attention pooling is computed using a weighted softmax function:

$$a = \text{softmax}\left(W^T H + b\right) \tag{4}$$

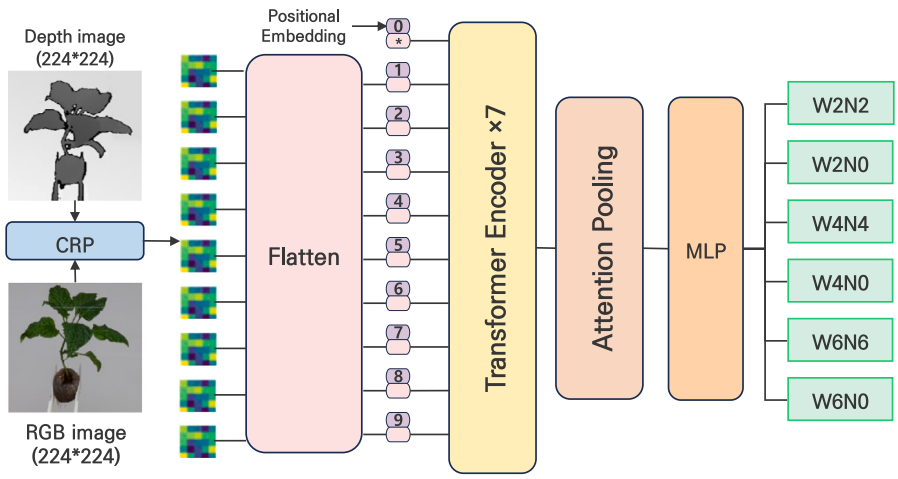


Fig. 6. RGBD-CTA-D architecture using data-level fusion by stacking RGB and depth channels.

Here, a denotes the attention weight, W is a learnable weight vector, and b is a bias term. This mechanism allows the model to dynamically assign an importance score to each feature in the sequence, thereby reflecting its contributions to the final classification decision.

Using these attention weights, the model computes a weighted average of the sequence features, generating a fixed-size output vector that is essential for the classification task. The weighted average is calculated as follows:

$$c = H^T a \tag{5}$$

Here, c represents the weighted context vector, providing the model with a compressed information-rich representation of the entire input sequence. This step converts variable-length input into a fixed-length output, making it suitable for input to the classifier.

Finally, the network employs a fully connected layer to process the weighted average feature vector c. This layer maps the comprehensive feature representation to the predicted classification label. The use of the fully connected layer further enhances the network's ability to capture crucial information and provides a highly flexible mechanism for adapting to different classification tasks. This design ensures that the network can generalize to various seedling stress classification scenarios and accurately identify different stress levels.

2.2.4. RGBD-CTA-D: Data-level fusion strategy

RGBD images combine the color, texture, and shape information from RGB images with the 3D structural information from depth images, providing a comprehensive visual representation that enables the model to understand the scene's content more comprehensively. The details captured by RGB images help the model visually identify different objects, while depth information enhances the model's understanding of the spatial position and shape of objects. This data fusion is particularly beneficial for improving the model's robustness under complex environmental conditions, such as changing lighting, occlusion, and visual noise (Garbouge et al., 2021).

Multimodal data fusion can occur at different levels, including the data level, feature level, and decision level (Sanchez-Riera et al., 2016). Data-level fusion treats different types of data as indistinguishable multi-channel inputs that are directly stacked at the model's input layer. Feature-level fusion considers the interrelationship between different modal data in the model's intermediate layers. Decision-level fusion independently processes each modal data at the model's output stage and then combines the results.

As shown in Fig. 6, we adopted a data layer fusion strategy, where RGB images and depth images are placed side by side as inputs to the model (Garbouge et al., 2021). This approach treats the depth image as the fourth channel of the RGB image, resulting in input data that includes both color and depth information. The advantage of this fusion strategy is that it requires minimal adjustments to the network structure. Only the input channels of the first convolutional layer need to be adjusted from 3 to 4, without significant changes to the subsequent network structure.

This basic modification enables the model to leverage depth information to complement color information, providing a richer data foundation for seedling stress classification. This strategy allows the model to learn from both color and depth information and provides the foundation for subsequent feature-level fusion, further improving the model's performance. A series of experiments validate the practical benefits of this approach in seedling stress classification tasks.

2.2.5. RGBD-CTA-F: Feature-level fusion via cross-attention

While the data layer fusion strategy provides convenience in model development due to its simplicity and minimal model modifications, it often fails to fully exploit and utilize the complementarity and correlations between different modalities (Li and Wu, 2024). To extract relevant features from different modalities and unique features learned from individual modalities, the following network structure is designed.

As shown in Fig. 7, we have designed four channels (RGB-CRP, Depth-CRP, Shallow-Fuse, and Deep-Fuse) in the tokenizer part to extract features from multimodal data. Two channels (RGB-CRP and Depth-CRP) independently process RGB and depth data, while the other two channels (Shallow-Fuse and Deep-Fuse) perform feature fusion through CA mechanisms. These layer structures provide the model with multi-scale features ranging from primary edge and color information to advanced object parts and components.

Due to the significant size variations among seedlings, multi-scale fusion was introduced when designing the CA channels. The concept of multi-scale representation has been validated (Li and Wu, 2024). Low-level feature maps have smaller receptive fields and larger scales, containing less semantic information and more low-level feature information, such as edges and colors. High-level feature maps contain more semantic information and advanced feature information, such as parts and overall structures of objects. CA channels leverage the advantages of different receptive fields to fuse RGBD information, allowing the network to adapt well to various multi-scale targets and improve classification accuracy.

CA plays a crucial role in effectively integrating information from different data sources such as RGB images and depth images in both spatial and channel dimensions, enhancing the quality of feature representation and improving the model's understanding of the data. By reducing the weight of unimportant features, the network can converge faster and improve computational efficiency, as shown in Fig. 8.

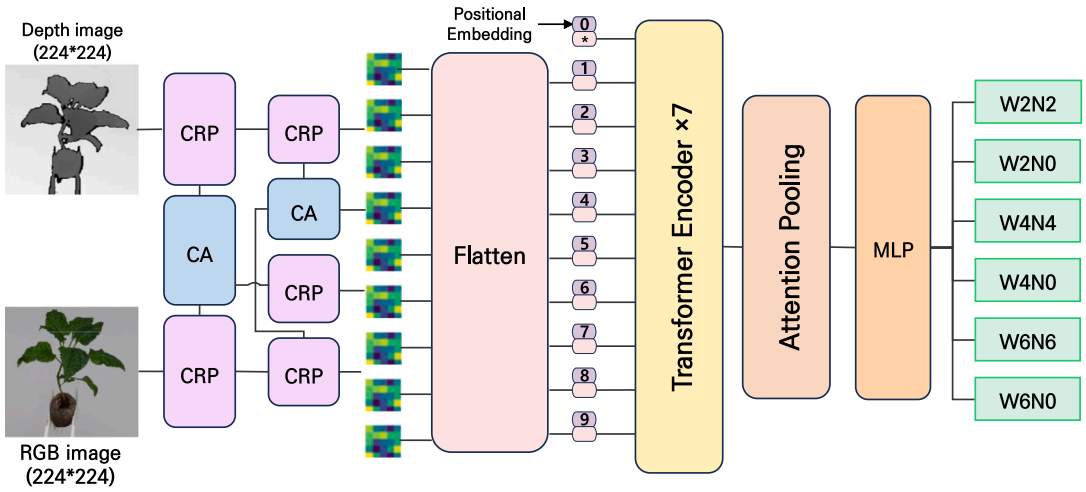


Fig. 7. RGBD-CTA-F architecture with CA module.

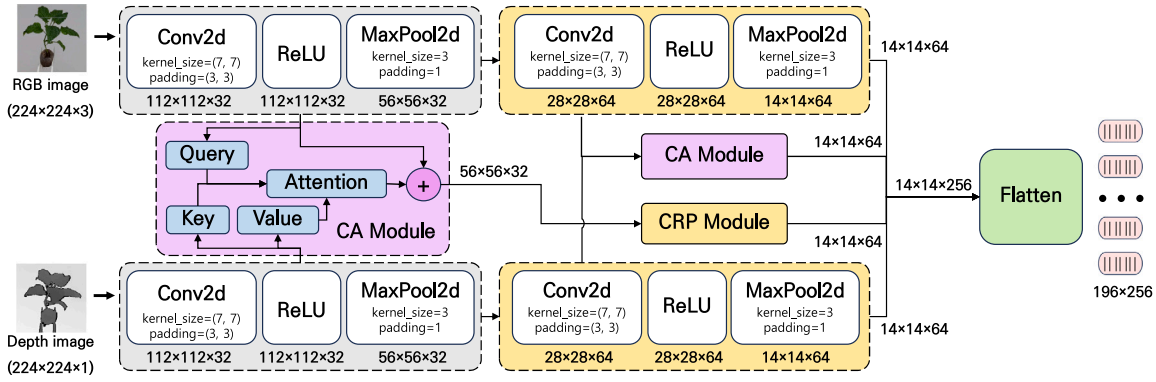


Fig. 8. CA module for multimodal RGBD feature integration.

The CA module integrates features from two separate CNN branches, one for depth image features and the other for RGB image features. These features are convolutional feature maps with the same spatial dimensions to facilitate CA calculations. A query–key–value mechanism is employed where RGB features serve as the query and depth features as key and value. The dot product between query and key is calculated, followed by a softmax function to obtain attention weights. These weights are then applied to the value to yield a weighted feature representation. Finally, the output of the CA is summed with the RGB features to achieve fusion, as follows:

$$F = \text{softmax}\left(\frac{(W^Q R)(W^K D)^T}{\sqrt{d}}\right) DW^V \tag{6}$$

$$C = F + R \tag{7}$$

where, R represents the RGB features, D represents the depth features. W^Q, W^K, and W^V are learnable lightweight matrices for query, key, and value, respectively. F is the weighted depth feature obtained by multiplying the attention scores with the depth feature values. C is the final fused feature, obtained by adding the attention mechanism's output F directly to the RGB feature R.

This fusion method not only preserves the original RGB features but also combines depth information weighted by the attention mechanism, allowing the network to enhance the performance of the classification task by learning appearance and scale information from different modalities. The implementation of this approach significantly improves the accuracy of the model in classifying seedling images, even in various environmental conditions, enabling robust identification of different stages of seedlings.

2.2.6. Training configuration

All training procedures were conducted on an embedded computing platform based on the NVIDIA Jetson AGX Xavier, operating under Ubuntu 20.04 with CUDA 11.4. Model development and training were implemented in python 3.8 using the pytorch 2.1 deep learning framework.

The training process was configured with a batch size of 32 and an initial learning rate of 5×10^{-4}, which was scheduled using cosine annealing with a minimum threshold of 1×10^{-5}. The optimizer adopted was AdamW, incorporating a weight decay coefficient of 6×10^{-2}. To stabilize convergence, a learning rate warm-up strategy was applied over the first 10 epochs starting from 1×10^{-6}, followed by 10 epochs of cooling and a total training duration of 200 epochs. Label smoothing with a smoothing factor of 0.1 was applied to mitigate overfitting and enhance model generalization. Additionally, automatic mixed precision was enabled to accelerate training and reduce memory consumption without compromising numerical stability.

To obtain a reliable estimate of generalization performance under limited data, we adopted an 80 %/20 % hold–out strategy combined with stratified 5-fold cross-validation on the training portion. Specifically, 20 % of the entire RGBD dataset was sequestered as an unseen test set and accessed only once for the final evaluation reported. The remaining 80 % was partitioned into five stratified folds that preserve the original class distribution. For each cross-validation iteration, four folds (≈ 64 % of the complete data) were used to train the network, and the remaining fold (≈ 16 %) served as a validation set for early stopping and hyper-parameter selection. All data splits were generated with a fixed random state to ensure reproducibility.

Table 3
Performance comparison of RGB-only, data-level RGBD fusion, and feature-level RGBD fusion methods.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Params (M)	GFLOPs	FPS
RGB-CTA	90.9	91.1	90.8	90.9	18.1	8.2	7.6
RGBD-CTA-D	91.2	91.4	91.1	91.2	18.4	8.6	7.4
RGBD-CTA-F	95.8	95.8	95.7	95.8	23.2	10.3	5.2

The classification objective was optimized using a cross-entropy loss function with label smoothing, where the smoothing factor $\epsilon = 0.1$. This strategy prevents overconfidence in model predictions and enhances generalization, particularly in cases where inter-class visual differences are subtle. The loss function is formally defined as:

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{c=1}^{C}\left[(1-\epsilon)\cdot y_{i,c} + \frac{\epsilon}{C}\right]\log(p_{i,c}) \tag{8}$$

where $y_{i,c}$ and $p_{i,c}$ denote the ground truth and predicted probability for class c in sample i, and C is the total number of classes.

3. Results and discussion

3.1. Performance analysis of fusion strategies

To comprehensively evaluate the impact of multimodal fusion strategies on seedling stress classification, we conducted comparative experiments on three network configurations: RGB-CTA, RGBD-CTA-D, and RGBD-CTA-F. Quantitative results are reported in Table 3. The FPS was measured on Jetson AGX Xavier using native PyTorch FP32 inference with a batch parameter of 1. The timing was averaged over 100 test images and included image preprocessing and inference threads running in parallel with robotic arm, and gripper control programs.

The performance metrics in Table 3 reveal the relative strengths of the three fusion strategies in terms of both overall correctness and class-level discriminability. RGB-CTA sets the baseline with a balanced yet restricted performance across all four metrics. This outcome reflects the information bottleneck that is intrinsically present in the single-modal RGB representation when differentiating subtle stress variations. The marginal improvements of RGBD-CTA-D over RGB-CTA indicate that simply appending depth as an extra channel offers limited complementary signals. It gives some noise reduction or contextual help but lacks an explicit way to integrate modality-specific semantics. In comparison with the baseline model, RGBD-CTA-F achieves approximately 5 % improvements in accuracy, precision, recall, and F1-score. These enhancements serve to validate the effectiveness of cross-attention-based feature-level fusion. By leveraging the complementary cues of RGB appearance and depth geometry, the network can more dependably distinguish subtle seedling stress patterns (Sun et al., 2022). The consistent precision and recall improvement indicates that RGBD-CTA-F enhances positive identification accuracy. It also improves the comprehensiveness of detecting stressed classes, thereby increasing the F1-Score. It is crucial for fine-grained multiclass stress classification where both false positives and false negatives carry operational cost.

Regarding computational efficiency and deployment feasibility, Table 3 reveals critical trade-offs between model complexity and performance. The RGB-CTA and RGBD-CTA-D models exhibit nearly identical parameter sizes (18.09 M vs. 18.35 M) and inference speeds (7.6 FPS vs. 7.4 FPS), reflecting their shared backbone architecture with only minor modifications to accommodate the depth channel. This finding indicates that merely adding depth as an additional input dimension incurs only a negligible computational burden. However, it also results in only minimal improvements in classification accuracy (from 90.9 % to 91.2 %). This observation underscores the restricted effectiveness of data-level fusion in harnessing multimodal synergy.

By contrast, the RGBD-CTA-F model, with its dedicated dual-stream encoder and CA fusion module, increases the parameter count to 23.22 M and reduces the inference speed to 5.2 FPS. While this represents a 28 % increase in model size and a 31.5 % decrease in speed relative to the baseline, the performance gain is substantial—boosting accuracy from 90.9 % to 95.8 %. The dual-stream encoders and cross-attention fusion introduce additional parameters and computation to better exploit complementary RGB and depth cues. Importantly, the model maintains a frame rate above the camera acquisition rate (5 FPS), ensuring it can process incoming frames in real time without introducing bottlenecks.

From a computational perspective, giga floating-point operations (GFLOPs) provide a more direct measure of workload than parameter count alone. The GFLOPs increase from 8.2 (RGB-CTA) to 8.6 (RGBD-CTA-D) and further to 10.3 (RGBD-CTA-F). This trend is consistent with the architectural evolution from a single-stream backbone to a dual-stream design with cross-attention-based feature fusion. While RGBD-CTA-D introduces only a minor increase in computation by expanding the input channel, RGBD-CTA-F incorporates parallel RGB and depth encoders as well as cross-attention operations, which substantially increase matrix multiplications and memory access operations. The observed decrease in FPS is therefore primarily attributed to the higher computational complexity and increased memory bandwidth demand. In particular, cross-attention modules introduce additional tensor reshaping and attention score computation.

Furthermore, given that the operating frequency of the transplanting robotic arm typically allows several hundred milliseconds for decision-making per seedling, the marginal latency introduced by RGBD-CTA-F is well within practical tolerances. These results indicate that the additional computational cost of feature-level fusion is acceptable, especially in precision agriculture scenarios where misclassification of stressed seedlings can incur significant downstream losses.

Since Table 3 cannot fully represent the predictions for each class, we further compared the performance of the three models using a confusion matrix, as shown in Fig. 9. According to the rules for plotting the confusion matrix, the numbers on the diagonal of each matrix (from the top-left to the bottom-right) represent the correct predictions for each class. The higher these numbers, the better the model's performance for the respective class. We analyze the confusion matrix generated by the RGB-CTA model on the test dataset, as shown in Fig. 9a, as the baseline. The RGB-CTA model displays significant off-diagonal dispersion, particularly in intermediate stress categories (e.g., W4N0 vs. W4N4 and W6N0), suggesting difficulty in distinguishing subtle visual differences.

The RGBD-CTA-D fusion approach treats depth and color equally, without according precedence to modality-specific semantics or dynamic relevance. A more detailed analysis of the confusion matrix reveals that, compared with the RGB-CTA baseline, the variety of error types has increased, as shown in Fig. 9b. Consequently, this approach results in information dilution rather than augmentation. Specifically, several intermediate stress categories (such as W2N0-W4N0) exhibit more frequent cross-class confusions. This implies that the incorporated depth channel introduces ambiguities that cannot be effectively resolved by simple concatenation. This discovery suggests that naive data-level fusion may inadvertently intensify subtle inter-class overlaps when the modalities are not adaptively coordinated. Thus, it underscores the need for more structured fusion mechanisms to clarify correlated yet distinct stress signatures.

In contrast, Fig. 9c reveals a predominantly diagonal distribution for RGBD-CTA-F, indicating the model's capacity to differentiate stress levels with high confidence. The success of this model can be attributed

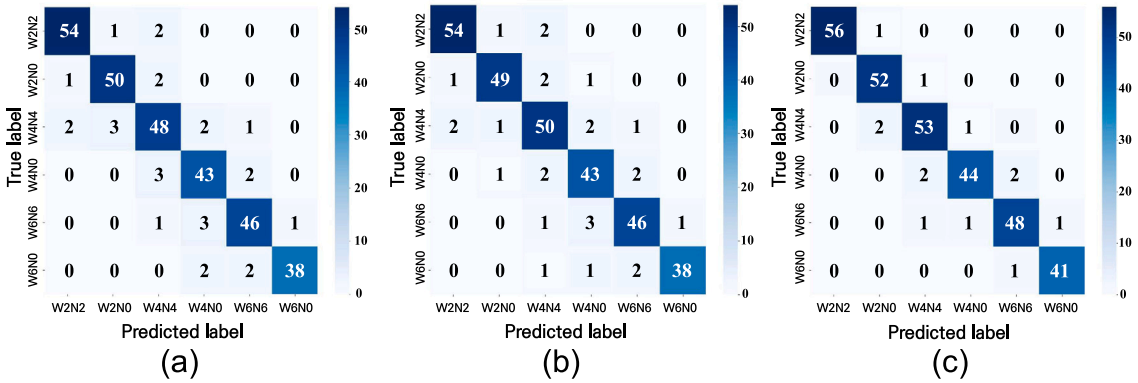


Fig. 9. Confusion matrix results: (a) RGB-CTA, (b) RGBD-CTA-D, and (c) RGBD-CTA-F on the test set.

to the integration of the CA module, which explicitly learns the relational dependencies between depth-induced structural cues and color-based visual patterns. By modeling RGB features as query vectors and depth features as keys and values, the network dynamically attends to spatial regions where structural deformation (e.g., curled leaves, compacted root systems) aligns with stress-induced discoloration, thereby enhancing class separability.

A closer examination of the confusion matrices indicates that most misclassified samples are concentrated in intermediate stress categories, such as W2N0, W4N4, and W4N0, where visual differences are subtle and class boundaries are inherently ambiguous. These hard-to-classify samples often share similar phenotypic characteristics, including slight chlorosis, moderate leaf shrinkage, and comparable canopy structure, making discrimination particularly challenging. In addition, variations in observation angle may cause partial leaf self-occlusion and canopy structure changes, occasionally masking discriminative symptoms such as localized chlorosis or wilted leaves. Moreover, forward–backward tilting of seedlings during robotic grasping can alter the camera-to-seedling distance, introducing inconsistencies in apparent scale that further increase classification ambiguity. RGBD-CTA-F reduces these confusions more effectively under occlusion and scale variation, indicating that cross-attention-based RGBD fusion improves robustness by jointly leveraging appearance and geometric cues.

In summary, the feature-level fusion architecture with CA not only amplifies the synergistic benefits of RGB and depth data but also facilitates semantic alignment across modalities, leading to noticeably higher classification accuracy and reliability. This validates the proposed model as a viable solution for real-time stress grading in precision seedling transplantation systems.

3.2. Interpretability via Grad-CAM attention maps

To further evaluate the interpretability and visual reasoning capabilities of the proposed models, we employed gradient-weighted class activation mapping (Grad-CAM) (Selvaraju et al., 2016) to generate class-discriminative localization maps. These visualizations help identify salient regions within the input image that contribute most significantly to the model's final prediction, thereby revealing the internal focus mechanism of different architectures.

Fig. 10 illustrates a series of Grad-CAM visualizations for random sample seedlings, highlighting the attention regions across different network variants. Warmer colors (e.g., red) indicate regions with higher contributions to the classification output, whereas cooler colors (e.g., blue) represent less influential or redundant areas. For each example, the predicted stress category and the corresponding confidence score are explicitly annotated, providing a direct visualization of the six-level stress classification outcome.

From the results in Fig. 10b, it is evident that the RGB-CTA model often concentrates on localized features such as root boundaries or individual leaves. These attention areas are narrow and fragmented. Although they occasionally align with stress indicators, they often fail to capture broader contextual or structural cues. This phenomenon accounts for the model's confusion at intermediate stress levels. The heatmaps also reveal that important areas, such as subtle chlorosis on multiple leaves or combined morphology cues, are not consistently aggregated. This results in unstable decision regions.

The RGBD-CTA-D variant in Fig. 10c shows more spatially extended attention compared to RGB-CTA, suggesting that the addition of depth information does help broaden the receptive focus. However, the attention is still irregular and exhibits scattered high-response patches that do not form semantically coherent regions. Further, it can be noted that the dispersed high-response patches did not effectively concentrate on the cues. This partial integration leads to an inconsistent weighting of appearance and geometric features. This phenomenon is manifested as both redundant and insufficient emphasis, which contributes to the heightened confusion among similar stress categories, as previously noted and as shown in Fig. 9b.

In contrast, the RGBD-CTA-F model Fig. 10d, which utilizes a CA-based feature fusion mechanism, demonstrates enhanced spatial awareness. Its attention regions are more uniformly distributed over semantically rich parts of the seedling, including leaf clusters, root collars, and inter-leaf spaces. Such patterns align closely with human-perceived stress indicators like leaf chlorosis, growth asymmetry, and shoot vigor. This suggests that the model is not only learning more relevant features but also reasoning more effectively by leveraging the complementary information from RGB and depth modalities.

Overall, the attention visualizations not only support the quantitative performance gains observed in previous sections but also demonstrate that the RGBD-CTA-F model exhibits satisfactory interpretability. This property is important for real-world agricultural deployment, where model transparency and trustworthiness are essential for user adoption and decision-making under variable field conditions.

3.3. Intra-dataset benchmarking against established lightweight models

In this section, we compare the proposed models with several open-source lightweight classification networks in our dataset. The goal of these experiments is to evaluate the deployment potential of our method on resource-constrained embedded devices, emphasizing both model size and classification accuracy. We chose models with parameter counts not exceeding 50 M to reflect the practical need for a balance between performance and compactness. The models were chosen to reflect a spectrum of design philosophies (efficient CNNs, hybrid/residual architectures, and transformer-derived models). A suitable model for real-world use must deliver high accuracy while remaining small enough to run efficiently on diverse edge platforms.

Table 4 presents a comparative analysis of the selected models. The proposed RGBD-CTA-F demonstrates outstanding performance across

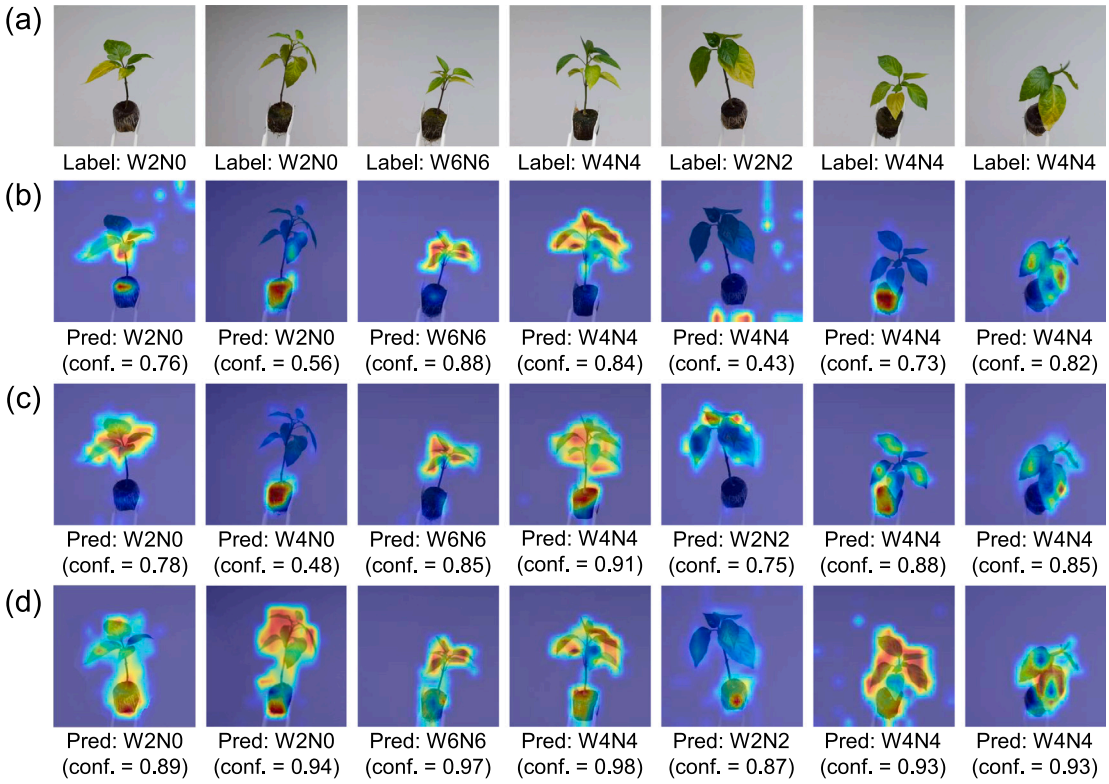


Fig. 10. Grad-CAM heatmaps highlighting salient regions for seedling classification with predicted stress categories (pred.) and confidence scores (conf.) across fusion methods: (a) original image with ground-truth label, (b) RGB-CTA, (c) RGBD-CTA-D, and (d) RGBD-CTA-F.

Table 4
Comparison of the proposed model with state-of-the-art lightweight classification networks.

Method	Acc (%)	Precision (%)	Recall (%)	F1-score (%)	Params (M)	GFLOPs	FPS
MobileNet v2 (Sandler et al., 2018)	82.7	83.1	82.5	82.8	6.9	0.6	35.6
SENet-29 (Hu et al., 2017)	83.2	83.2	83.5	83.4	35.0	3.8	14.3
ResNeSt-101 (Zhang et al., 2022)	93.5	93.2	93.6	93.4	48.0	16.3	3.4
TResNet-M (Ridnik et al., 2020)	91.1	91.5	90.8	91.2	29.4	4.8	11.9
ShuffleNet v2 (Ma et al., 2018)	85.6	85.8	85.1	85.5	15.6	1.3	29.0
Swin-T (Liu et al., 2021)	79.5	78.1	78.6	78.4	28.0	4.5	8.8
CvT-13 (Wu et al., 2021)	76.3	75.3	76.5	75.9	19.6	6.7	7.0
ResNet50-S (He et al., 2015)	80.9	80.4	80.6	80.5	22.1	4.1	13.3
Swin-S (Liu et al., 2021)	88.2	88.1	88.5	88.3	50.0	8.7	5.1
RGB-CTA	90.9	91.1	90.8	90.9	18.1	8.2	7.6
RGBD-CTA-D	91.2	91.4	91.1	91.2	18.4	8.6	7.4
RGBD-CTA-F	95.8	95.8	95.7	95.8	23.2	10.3	5.2

all metrics. Furthermore, a larger number of model parameters does not necessarily lead to higher accuracy. For instance, MobileNet v2 achieved an accuracy of 82.7 % with only 6.9 M parameters, highlighting its strong parameter efficiency. However, ResNet50-S has a significantly larger size (22.1 M parameters), attained the accuracy of 80.9 % among the compared models. This suggests that while larger models perform better in some cases, efficient architectures can still achieve competitive results with far fewer parameters. Despite slightly higher GFLOPs compared with some ultra-lightweight models, the proposed architecture maintains a practically deployable FPS while achieving substantially higher classification accuracy, indicating a favorable balance between inference speed and predictive performance.

To further visually compare the trade-off between accuracy and model size more intuitively, Fig. 11 plots the data as a 2D scatter plot. A weak positive correlation (the red dotted least-squares trend line) is clearly observable: as models increase in size, accuracy generally shows an upward tendency, although the slope remains relatively shallow. Based on the trend line, the scattered points separate into three strata. Firstly, models fall well below the trend line (such as CvT-13 and Swin-T) are primarily based on transformer architecture. They struggle to adapt to small sample datasets and are difficult to train effectively

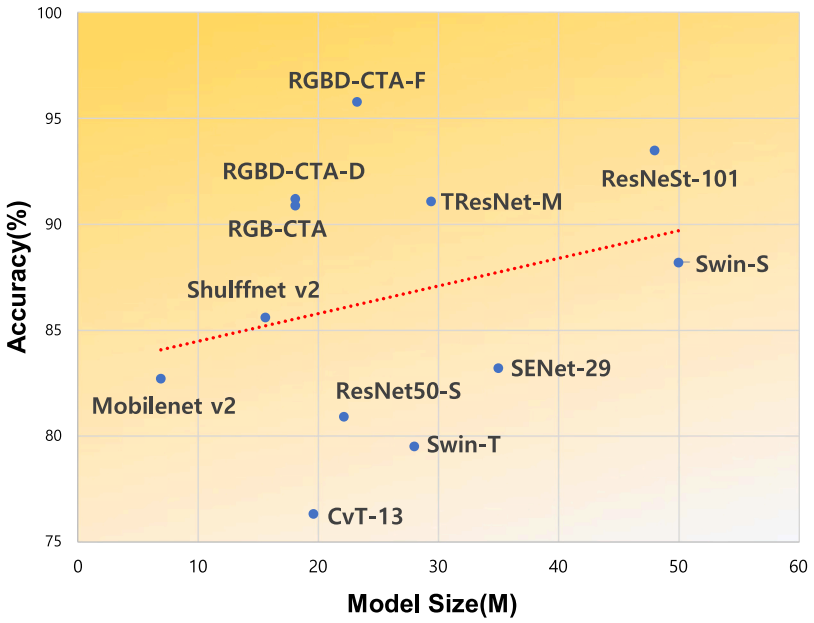


Fig. 11. Comparison of classification accuracy versus model size across lightweight architectures.

Table 5
Comparison between reported studies and the proposed model.

Study	Data pieces	Class	Model	F1 score (%)
Otoya and Gardini (2021)	714	4	LinearSVC	74.0
Zhao et al. (2023)	1730	5	YOLO-ShuffleNet	90.2
RGBD-CTA-F	1537	6	CNN–Transformer	95.8

without local convolution or hybrid gating mechanisms. A middle group, exemplified by MobileNet v2, ShuffleNet v2, and ResNeSt-101, lies in close proximity to the regression line. Consequently, it achieves an accuracy that is proportional to the model size, yet not superior to it. Finally, our approach attains precise classification accuracy while sustaining a moderate scale. Cross-modal feature alignment offers a higher information density within a restricted parameter budget. RGBD-CTA-F improves accuracy by 2.3 % over the much larger ResNeSt-101 while requiring only half the parameters that directly translates into lower inference latency and power consumption on edge hardware.

The comparison demonstrates that carefully designed multimodal fusion can outperform more generic or heavier architectures by extracting synergistic information efficiently, avoiding the diminishing returns observed in over-parameterized or poorly adapted models.

3.4. Cross-study on seedling state classification

The performance of the proposed model was compared with that of the published literature, which presented its findings in the domain of seedling state classification. Table 5 offers a comprehensive comparison of these models. This comparison takes into consideration the number of datasets and categories employed, along with the F1 score.

Table 5 contrasts the proposed network with two representative studies on seedling grading. Otoya and Gardini (2021) addressed artichoke seedlings using hand-crafted features and a shallow Linear-SVC classifier. Despite the task involved only four classes and a modest sample size (714 images), the F1 score reached just 74.0 %, highlighting the limitations of traditional feature engineering when intra-class variability is high. Zhao et al. (2023) introduced an RGB, side-view YOLOv5s variant equipped with a ShuffleNet-V2 backbone. On a larger five-class dataset (1730 images) the detector achieved a markedly higher F1 of 90.2 %, confirming the benefit of end-to-end feature learning and multi-indicator detection during transplanting. Since the above-mentioned literature did not directly provide the F1 score, this study calculated it using the parameter transformation method. This result serves as a rough indicator for cross-literature comparison, and the actual value have a deviation of 1–5 %.

Against these baselines, RGBD-CTA-F model attains an outstanding performance, delivering a F1 score of 95.8 % on a six-class dataset (1537 images). Certainly, direct numerical comparison should be interpreted with caution because the datasets differ in crop, imaging geometry, and class definitions. Despite these differences, the trend in Table 5 indicates a clear progression: moving from handcrafted to deep RGB, and finally to RGBD with feature-level fusion consistently elevates grading accuracy.

3.5. Ablation study: Effect of attention pooling module

The classifier part of the proposed method utilizes the attention pooling module, which is an effective mechanism for enhancing the model's ability to process long sequential data. Attention pooling can aggregate information from the input sequence, providing a compressed information representation. It is particularly important for dealing with limited attention spans. To validate the efficacy of this design, in this section, ablation studies are carried out on the prediction module. Specifically, the attention pooling module is removed from the classifier section, and the variations in the model's performance are meticulously observed.

From Table 6, it can be observed that the removal of the attention pooling module resulted in performance degradation across different model variants. The performance of the RGB-CTA model decreased by approximately 2 %, RGBD-CTA-D decreased by approximately 3.5 %, and RGBD-CTA-F experienced the smallest decrease, at only around 1 %. This slight performance drop be because the RGBD-CTA-F model was already quite effective in data fusion using the CA module, so even when the attention pooling module was removed, its performance was relatively less affected. The GFLOPs and FPS remain nearly unchanged after removing the attention pooling module, as its computational

Table 6
Ablation results of removing the attention-pooling module.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	GFLOPs	FPS
RGB-CTA	90.9	91.1	90.8	90.9	8.2	7.6
RGB-CTA[a]	88.5	89.8	88.5	88.4	8.2	7.6
RGBD-CTA-D	91.2	91.4	91.1	91.2	8.6	7.4
RGBD-CTA-D[a]	87.7	88.9	87.8	87.6	8.6	7.4
RGBD-CTA-F	95.8	95.8	95.7	95.8	10.3	5.2
RGBD-CTA-F[a]	94.7	94.6	94.6	94.6	10.3	5.2

[a] Removal of attention-pooling module.

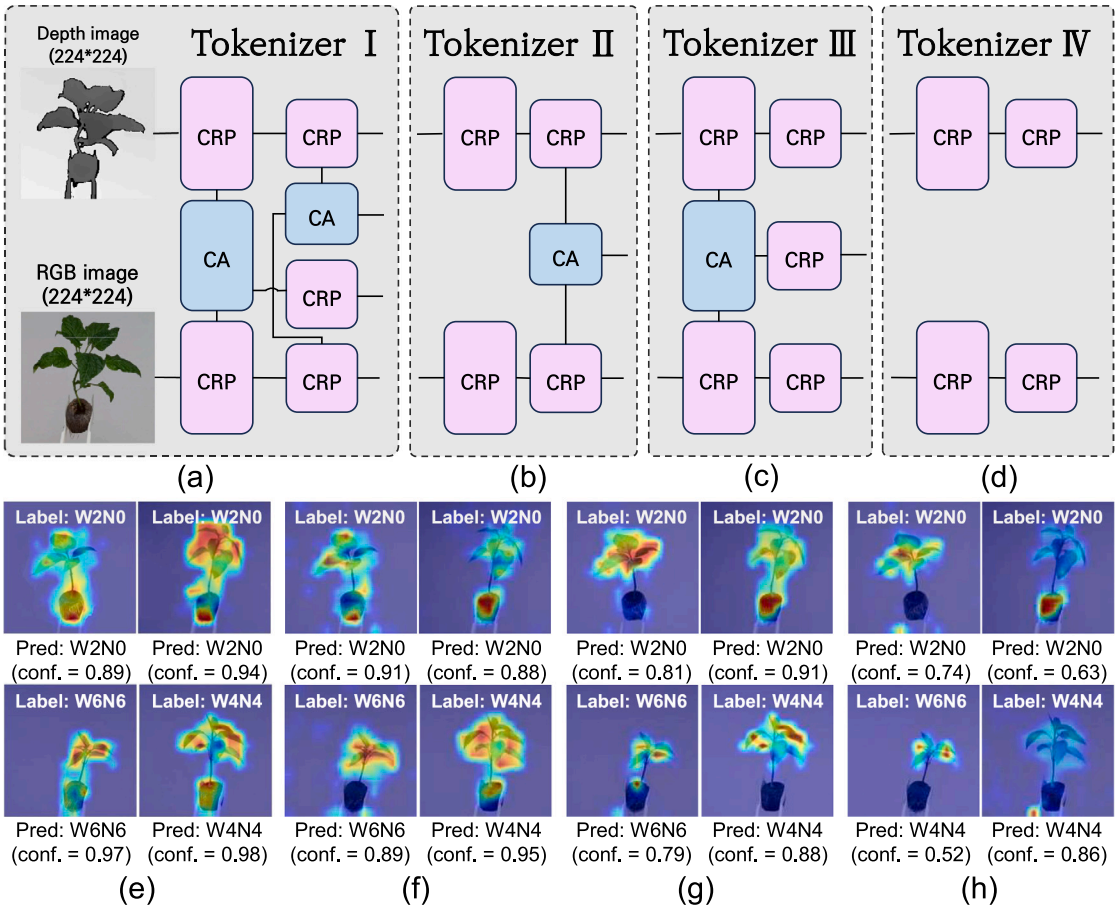


Fig. 12. Tokenizer designs and Grad-CAM visualizations: (a) Tokenizer I: original model, (b) Tokenizer II: removal of shallow feature fusion module, (c) Tokenizer III: removal of deep feature fusion module, (d) Tokenizer IV: removal of all fusion modules, (e–h) Grad-CAM visualization examples for Tokenizer I–IV, respectively.

cost is negligible (0.0004 GFLOPs) compared with the transformer encoder, indicating that this component mainly contributes to feature aggregation and classification performance rather than computational complexity. This emphasizes the importance of using the attention pooling module in the classifier section to improve the overall detection performance of the model.

3.6. Ablation study: Effectiveness of cross-attention fusion

To rigorously evaluate the impact of feature fusion hierarchy in the tokenizer, we designed four ablation variants (tokenizer I–IV) with progressively reduced fusion complexity, as illustrated in Fig. 12. Tokenizer I (Fig. 12a) integrates both shallow and deep-level RGBD features via CA modules, enabling multi-scale fusion. Tokenizer II and III ablate shallow and deep fusion paths respectively, while tokenizer IV serves as a fusion-free baseline. All the ablation models are guaranteed to have an output of $14 \times 14 \times 256$ through a 1×1 linear projection, ensuring consistency with the input interface of the complete model.

As shown in Table 7, tokenizer I achieved the highest performance (95.8 %), indicating that its strategy of fusing shallow and deep features is effective. In deep learning, deep network features tend to be more abstract and can capture high-level semantic information, and tokenizer I can leverage these features to improve the overall performance of the model. While tokenizer II's performance (93.0 %) is slightly lower than tokenizer I, it still outperforms tokenizer III (91.9 %), which only utilize features from shallow fusion module. This further emphasizes the importance of deep features in model performance. In contrast, tokenizer IV lacks the CA mechanism and exhibits the weakest performance (89.6 %). This indicates that the CA mechanism enhances the model's ability to capture key features by focusing on important features and suppressing irrelevant information. The GFLOPs and FPS values also reveal that the additional cross-attention fusion introduces only moderate computational overhead while maintaining a practically usable inference throughput.

To further investigate how different modules influence the model's decision process, Grad-CAM visualizations are provided for the four tokenizer variants. As shown in Fig. 12e, tokenizer I exhibits a more

Table 7
Performance of the Tokenizer designs with varying feature fusion hierarchies.

Type	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	GFLOPs	FPS
Tokenizer I	95.8	95.8	95.7	95.8	10.3	5.2
Tokenizer II	93.0	93.1	92.9	92.9	9.4	5.7
Tokenizer III	91.9	92.3	91.7	91.8	10.1	5.4
Tokenizer IV	89.6	90.3	89.5	89.6	9.1	5.9

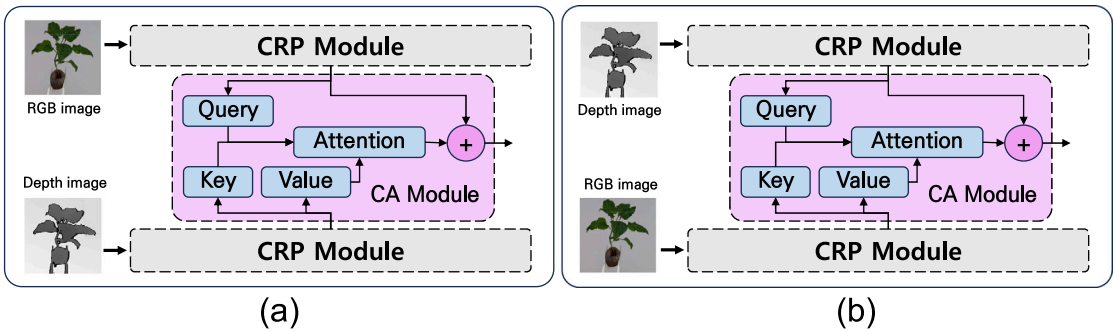


Fig. 13. Fusion direction: (a) RGBD: original model, and (b) DRGB: swap RGB and depth input channels.

Table 8
Impact of fusion direction in the CA module on classification performance.

Type	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
RGBD	95.8	95.8	95.7	95.8
DRGB	92.2	92.8	91.9	92.0

globally distributed yet semantically focused activation pattern across the entire plant structure, highlighting the most informative regions while suppressing irrelevant background responses. In contrast, the ablated models exhibit more scattered or incomplete attention patterns, particularly when either shallow or deep fusion is removed, as shown in Fig. 12f–h. These observations provide intuitive evidence that cross-attention fusion improves the network's ability to capture discriminative stress features from RGBD inputs.

These results validate that hierarchical CA is essential for maximizing the complementary strengths of RGB and depth modalities. Shallow fusion can capture coarse structural cues such as leaf edges and contours, while deep fusion can focus on more detailed information. Their combined effect fosters robust multimodal feature learning, enhancing stress classification under real-world variability.

3.7. Ablation study: Impact of fusion direction in CA module

In the design of the CA module, we selected RGB features as the query (primary feature) and depth features as the key and value (reference features). This is a common and effective design pattern in multimodal learning. This pattern allows the model to leverage depth features to enhance and complement the information provided by RGB features. In this design, depth features are not simply parallel to RGB features; instead, they interact and fuse meaningfully with RGB features through the attention mechanism. To validate the effectiveness of this design, ablation studies were conducted, and the network structure is as shown in Fig. 13.

As shown in Table 8, the RGBD configuration achieves satisfactory performance across all metrics, with an accuracy of 95.8 % and an F1-score of 95.8 %, whereas the reversed DRGB variant exhibits a notable performance drop (accuracy: 92.2 %, F1-score: 92.0 %). The performance gap highlights a critical insight into modality prioritization in CA design: employing RGB features as queries yields more effective attention guidance than the reversed configuration. RGB images inherently encapsulate rich semantic cues — such as color gradients, stress-induced discoloration, and textural variations — that are highly discriminative for fine-grained stress level classification. In contrast, depth data, while useful for conveying structural geometry and spatial morphology, lack the visual semantics necessary for precise query formulation. When RGB features serve as queries and depth features act as keys and values (RGBD configuration), the network learns to attend selectively to spatial regions where depth cues reinforce or contextualize color-based anomalies. Conversely, using depth as the query (DRGB configuration) introduces semantic misalignment, as low-level geometric features struggle to effectively interrogate high-level appearance features in the RGB modality. This misdirection compromises feature integration and degrades classification accuracy, underscoring the importance of modality-aware role assignment in cross-modal attention mechanisms.

This finding underscores the importance of query–key asymmetry in multimodal attention design. Assigning semantically richer features (e.g., RGB) as the query source enhances the network's capacity to extract context-aware correlations from auxiliary modalities (e.g., depth). This improves robustness and generalization in seedling stress classification tasks.

4. Limitations and future work

The CNN–Transformer-based algorithm was proposed in this study for the classification of seedling stress levels, incorporating a lightweight CA mechanism to enable efficient fusion of RGBD multimodal data. Although the proposed method demonstrated competitive performance in terms of classification accuracy, real-time processing, and embedded deployment, several limitations remain that warrant further investigation for broader applicability and technical advancement.

The model's generalization capacity is limited by the scale and diversity of the current training dataset, which primarily includes a single crop species (bell pepper) under standardized stress conditions with limited variability in seedling age and environmental context. Such a narrow distribution raises concerns about domain shift, potentially leading to performance degradation when applied to other crop types, natural field environments, or multi-source stress factors. Future work should therefore focus on constructing a more representative RGBD dataset encompassing multiple crop species, growth stages, and diverse

environmental settings to improve the robustness and generalizability of the model in real-world scenarios.

Although the model has been designed with lightweight components for edge deployment, further optimization is feasible and necessary. Advanced model compression techniques such as pruning and quantization, along with efficient convolutional architectures (e.g., depthwise separable convolutions), could be employed to reduce inference latency and computational cost. Moreover, leveraging dedicated hardware accelerators such as neural processing units (NPU) would facilitate real-time and energy-efficient inference, thereby lowering the practical deployment threshold and enhancing applicability in agricultural equipment with limited resources.

In summary, future research should focus on improving data diversity, developing automated and integrated preprocessing workflows, optimizing network efficiency through model compression, enhancing cross-scenario generalization, and ensuring adaptability to hardware deployment. These advancements could facilitate practical implementation and scalability of the proposed visual system in intelligent agriculture applications.

5. Conclusion

In this study, a lightweight CNN–Transformer framework was developed to assess seedling stress in real time during robotic transplanting by integrating color texture and depth geometry through a dual-stream CA strategy. The network employs a multi-scale tokenizer for rapid feature extraction, transformer encoders for long-range context, and attention pooling for compact classification. Model performance was evaluated on a dedicated RGBD dataset comprising seedlings subjected to graduated water-nutrient stress treatments and was benchmarked against several compact vision networks as well as a channel-stacking baseline. Experimental results demonstrate that the proposed RGBD-CTA-F model achieves an overall classification accuracy of 95.8 % on a six-level stress dataset, while sustaining a real-time inference speed of 5.2 FPS. Qualitative and statistical comparisons showed that the proposed architecture delivered the most reliable stress-level predictions while keeping computational requirements within the limits of an embedded Jetson platform. Future work will incorporate model compression for ultra-low-power devices and cross-crop validation throughout the leafy growth period, thereby enabling continuous vitality monitoring and more responsive nursery management.

CRediT authorship contribution statement

Yu Zheng: Writing – original draft, Visualization, Formal analysis, Data curation. **Seong-Min Kang:** Writing – original draft, Investigation, Data curation. **Do-Hwan Kim:** Methodology, Investigation. **Yunde Shen:** Validation, Supervision. **Kyeong-Hwan Lee:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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